DOCUMENT 08

Hardware Specification Compendium

All Drone Units · Blueprints · Wiring Guides · Assembly Procedures · Maintenance

Complete engineering reference for every hardware unit in the Modular UAV Fleet Ecosystem. Covers seven unit types: two worker variants (W1 stripped, W2 full-payload), command node (C1), two carrier platforms (K1 ground, K2 airborne), and two docking station variants (D1 fixed, D2 carrier-mounted). Each section contains physical specifications, full component list with part numbers, wiring/connection diagrams, software configuration guide, and maintenance schedule.

Project:	Modular Autonomous UAV Fleet Ecosystem
Document Reference:	UAV-DOC-08-HW-2026
Estimated Pages:	30
Author:	Shrey Kumar — SRM IST Chennai
Classification:	Confidential Research Document
Version:	1.0 — May 2026

UNIT TAXONOMY AND DESIGNATION SYSTEM

The fleet uses a structured unit designation: the first letter identifies tier (W=Worker, C=Command, K=Carrier, D=Dock), the number identifies the variant within tier. Each unit is assigned a unique serial number in the format **UAV-[CODE]-[YYMMDD]-[NNN]** at assembly time and logged to the fleet database.

Unit	Full Name	Tier	Role	Qty (Phase 1)	Qty (Phase 2)
W1	Worker Drone — Stripped	1	Maximum-endurance execution	0 (planned)	3
W2	Worker Drone — Full Payload	1	Validated execution with sensors	2	2
C1	Command Node	2	AI inference, fleet coordination, relay	1	1
K1	Ground Carrier Platform	3	Battery logistics, dock, comms relay	1	1
K2	Airborne VTOL Carrier	3	Mobile logistics, extended range	0	1
D1	Fixed Docking Station	3	Ground recharge and alignment	1	2
D2	Carrier-Mounted Dock	3	Integrated into K1/K2	2 bays in K1	4 bays

Table 1. Unit taxonomy and deployment quantities by phase.

W1 Worker Drone — Stripped Configuration

Tier: Tier 1
Mass: 1.65 kg MTOW
Endurance: ~159 min predicted

W1 is the endurance-optimised worker drone designed according to the battery mass fraction optimality theorem (M_b* = 2×M_f). By removing all AI compute, LiDAR, and depth camera, fixed subsystem mass is reduced to 0.55 kg, enabling a 1.10 kg Li-ion battery at the optimal 2:1 battery-to-fixed mass ratio. W1 carries no onboard intelligence; all navigation is provided via GPS injection from the command node or federated EKF via UWB.

W1 Physical Specifications

Parameter	Value	Notes
MTOW	1.65 kg	Battery 1.10 kg + fixed 0.55 kg
Frame diagonal	650 mm	Tarot TL65B44 motor-to-motor
Frame dimensions (body)	~195 × 195 × 55 mm	H excl. props
Prop diameter	228.6 mm (9 in)	HQProp 9×4.5 3-blade carbon
Total disc area	0.1642 m²	4 × π × (114.3 mm)²
Predicted hover power	41.9 W (prop) + 4.0 W (avionics)	k_emp × (1.65 × 9.81)^1.5
Predicted endurance (hover)	159 min	Eq. (6) core paper — not yet flown
Max forward speed	~18 m/s	Estimated from W2 validated data
IP rating	IP54 minimum	Sealed connectors; conformal coated PCBs
Operating temperature	0°C to 45°C	Battery pre-heat below 10°C
GPS-denied navigation	Federated EKF via UWB	DWM3000; σ_pos ≤ 0.20 m at 60 s
Docking method	IR beacon + magnetic keel	D1/D2 compatible; 500 Hz guidance

Table W1-1. Physical specifications.

W1 Component List

#	Component	Part Number / Spec	Qty	Mass (g)	Cost ($)
01	CFRP Frame Set	Tarot TL65B44	1	320	85
02	Brushless Motor	T-MOTOR F90-1300KV	4	144	152
03	FOC ESC	T-MOTOR F55A Pro II	4	160	288
04	Propeller (CW+CCW)	HQProp 9×4.5×3 carbon	4+4 spare	36	16
05	Flight Controller	Holybro Pixhawk 6C	1	60	110
06	GPS + Compass	Holybro M9N u-blox	1	36	45
07	UWB Module	Decawave DWM3000	1	10	35
08	Li-ion Battery	4S 10,000 mAh 18650 ~148 Wh	1	1100	145
09	Power Module	Holybro PM02D (isolated)	1	12	29
10	LoRa Telemetry	Holybro SiK 915 MHz	1	15	35
11	ELRS Receiver	ExpressLRS EP2 2.4 GHz	1	5	18

#	Component	Part Number / Spec	Qty	Mass (g)	Cost ($)
12	IR Dock Receiver PCB	Custom: 4× TEMD6010 + STM32G0	1	12	28
13	Docking Keel Rail	Custom CNC Al 6061-T6 anodised	1	55	45
14	NdFeB Magnets	N52 10×3 mm 4 kg pull	4	8	8
15	Pogo Pins (power)	Gold-plated 10A spring ×4	4	6	12
16	Custom PDB	ESC dist + 5V isolated + sensing	1	28	45
17	ESC Heatsinks	Cu 40×20×10 mm	4	32	12
18	Vibration Isolators	M3 rubber grommet	8	8	5
19	Wiring + Connectors	XT60/XT90, 12AWG silicone	1 set	35	18
20	Fasteners	M2/M3 titanium + threadlock	1 set	12	12
	TOTAL			**~2,094 g** (excl. props)	**~$1,147**

Table W1-2. Complete component list. Target MTOW 1,650 g — frame mass distribution: battery 1,100 g; fixed subsystems 550 g.

W1 Wiring Block Diagram

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ 4S Li-ion 1.10 kg ■
■ (XT90 main connector) ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ VBAT (~14.8V)
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ ■ ■
■■■■■■■▼■■■■■■■■■ ■■■■■■■■■■■■▼■■■■■■■■■■■ ■■■■■■■■■■■■▼■■■■■■■■■■■■■
■ Custom PDB ■ ■ 4× FOC ESC ■ ■ Custom PDB ■
■ (isolated) ■ ■ T-MOTOR F55A ■ ■ current sensor ■
■ 5V/4A rail ■ ■ + heatsink ■ ■ → Pixhawk ADC ■
■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■■
■ ■ PWM+DShot
■ ■■■■■■■■■■■■■■■■■■■■■■■■
■ ■ 4× T-MOTOR F90 ■
■ ■ + HQProp 9×4.5 ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■
■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Holybro Pixhawk 6C ■
■ IMU: ICM-42688-P × 2 ■ Baro: ICP-20100 ■
■ UART1: LoRa SiK ■ UART2: GPS M9N ■
■ UART3: ELRS RX ■ SPI: DWM3000 UWB ■
■ CAN: Custom PDB ADC ■ GPIO: IR board trigger ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ UART (parsed)
■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ IR Docking Receiver PCB (belly-mounted) ■
■ 4× TEMD6010 photodetectors + STM32G031 ■
■ Output: [ε_x, ε_y, h] at 500 Hz over UART ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

W1 Top-View Layout

```
FRONT
[Motor FL]
/ \
```

```
[Motor RL] [Motor FR]
\ /
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ GPS UWB ■ ← top deck
■ [Pixhawk6C] ■
■ [PDB+regs] ■
■ [IR PCB ↓] ■ ← belly mount
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
/ \
[Motor LL] [Motor LR]
\ /
[Motor RL]
REAR

Battery: under main frame, secured with velcro + strap
Docking keel: ~80 mm rail centred on belly longitudinal axis
CG target: ±5 mm of motor plane centroid
```

W1 Software Configuration (PX4 / QGroundControl)

- **Airframe:** Generic Quadcopter X (QGC airframe selector)
- **Motor outputs:** ESC calibration via BLHeli32 Suite; verify CW/CCW assignment before first prop-on test
- **Power module:** Holybro PM02D; calibrate voltage divider and current sensor in QGC Power Setup
- **Battery:** 4S 10,000 mAh; voltage levels: full=16.8V, land=14.0V, RTH trigger=14.4V (≈40% SOC)
- **GPS:** u-blox M9N on UART2; set GPS_TYPE=1, COMPASS_DEV_ID match for M9N
- **UWB:** DWM3000 on SPI2; enable UWB_ENABLE parameter (custom PX4 fork); set NODE_ID unique per drone
- **IR board:** STM32G031 UART at 500 Hz; PX4 custom driver parses [ε_x, ε_y, h] and maps to LANDING_TARGET message
- **RTH altitude:** 25 m AGL minimum; configured via RTL_RETURN_ALT
- **Battery failsafe:** BATT_LOW_THR=40% (RTH); BATT_CRIT_THR=25% (land in place); BATT_EMERGEN_THR=15% (immediate land)

W1 Pre-Flight Checklist

- ■ Battery voltage ≥16.0V (90%+ SOC); check cell balance ≤50 mV spread
- ■ All 4 props installed and hand-torqued; no cracks; no nicks at leading edge
- ■ GPS lock ≥8 satellites; HDOP ≤1.5; home position set
- ■ ELRS link confirmed; all channels responding; failsafe triggers correct mode
- ■ LoRa telemetry connected; GCS receiving heartbeat
- ■ UWB ranging confirmed with at least one peer or beacon
- ■ IR board self-test: send UART query, confirm 4-channel demodulation active
- ■ Pogo pins clean; no debris on keel rail contacts
- ■ Motors spin test (no props): all CW/CCW correct; no abnormal current draw
- ■ IMU vibration check: hover on stand <5 s; confirm RMS ≤0.5 mm/s in QGC

W1 Maintenance Schedule

Interval	Task
Every flight	Visual inspect frame, props, connectors; check battery cell balance
Every 5 flights	Clean pogo pins with IPA; inspect keel rail for deformation; check all screw torque
Every 20 flights	Check motor bearing smoothness by hand; inspect ESC heatsink adhesion; calibrate compass
Every 50 flights	Recalibrate IR board zero offset; inspect CFRP arms for micro-cracks; replace pogo pin springs

Interval	Task
Every 200 hours	Replace motor bearings; replace propellers; inspect battery internal resistance (target <15 mΩ/cell)
Every 300 cycles	Replace pogo pins; inspect keel rail surface for galvanic corrosion; re-apply conformal coat PCBs

Table W1-3. Maintenance schedule.

W2 Worker Drone — Full Payload (Validated)

Tier: Tier 1
Mass: 1.97 kg MTOW
Endurance: 38 min validated

W2 is the currently validated test platform. It carries the Jetson Orin NX companion computer, Intel RealSense D435i, and a 74 Wh 4S LiPo battery. All core paper measurements — FOC acoustics, CFRP FEA, SPF navigation, power isolation, and endurance data — were obtained on this configuration. W2 serves as the command node proxy during Phase 1 and as a worker with onboard sensing capability for missions requiring edge inference.

W2 Physical Specifications

Parameter	Value	Validation Status
MTOW	1.97 kg (470 g full payload)	Validated — measured
Hover power (prop)	54.28 ± 1.0 W	Validated — n=10 trials
Hover power (total)	83.1 W (incl. full AI pipeline)	Validated — measured
Hover endurance	38.0 ± 2.1 min at 71.1% DOD	Validated — measured
FOC noise at 5,000 RPM	55.0 dB SPL @ 1 m	Validated — anechoic chamber
FOC vibration reduction	11.3× vs trapezoidal	Validated — strain gauges
CFRP safety factor	4.83 under 3g load	Validated — ANSYS FEA
SOC estimation (CFRP)	0.9% RMS error	Validated — EKF bench
Brownout rate (with PM02D)	0%	Validated — 30+ flights
AI navigation success (SPF)	92.7% (15 real-world trials)	Validated — indoor course
Battery	4S 5,000 mAh LiPo (74 Wh, 385 g)	Validated
Frame	Tarot TL65B44 CFRP, 650 mm diagonal	Validated

Table W2-1. W2 specifications with validation status.

W2 Additional Components vs W1

Component	Spec	Mass (g)	Power Draw	Cost ($)
NVIDIA Jetson Orin NX 16GB	100 TOPS, 25W TDP, 100 TOPS	158	4.1–26.1 W	499
Holybro Pixhawk-Jetson Baseboard	Isolated 5V/4A PSU, CAN, UART	45	Included in PM02D	95
Heatsink (passive)	40×40 mm 8-fin Al, 5 W/m·K TIM	28	0	15
Active fan (hot ambient)	40 mm 5V 0.3W axial	12	0.3 W	8
Intel RealSense D435i	RGB-D 30fps, 0.1–10 m, 72 g	72	1.5–2.5 W	179
256 GB microSD	Industrial, UHS-II	2	0.1 W	45
4S LiPo 5,000 mAh	Tattu 45C (replaces W1 Li-ion)	385	—	55

Table W2-2. Additional W2 components. W2 total parts cost ~$1,310.

W2 Thermal Management Notes

- Jetson Orin NX throttles at T_junction = 95°C. At 25°C ambient: junction ≈ 63°C under full VLA pipeline — passive heatsink sufficient.
- At 35°C ambient (Chennai summer): junction ≈ 73°C — passive marginal; activate 40 mm fan.
- At 40°C ambient: junction ≈ 78°C with fan — acceptable. Above 45°C: limit AI to YOLOv8n only (12.4 W instead of 24.8 W).
- ESC temperature: 58°C at 5,000 RPM FOC with heatsink. At 40°C ambient: ~73°C — within 80°C limit but monitor.
- Energy-aware mode: when SOC < 40%, resolution drops 640→320px; GPU power drops 12.4→7.1 W; endurance +11.4%.

C1 Command Node

Tier: Tier 2
Mass: ~2.4 kg MTOW
Endurance: 35–45 min (estimated)

C1 is the fleet intelligence hub. It carries the Jetson Orin NX, Livox Mid-360 LiDAR, Intel RealSense D435i, dual-antenna RTK GPS for precise heading, and the fleet coordination radio stack. C1 accepts the weight and power penalty of the full sensor suite in exchange for fleet-level capabilities: cooperative localisation of GPS-less workers, CBBA-SOC task assignment, AI inference for all workers, and long-range GCS relay. C1 remains airborne as an elevated intelligence platform, not a delivery vehicle.

C1 Physical Specifications

Parameter	Value	Notes
MTOW	~2.4 kg (payload ~900 g)	Estimated — heavier than W2
Frame	Tarot TL65B44 modified or custom larger frame	May require 700 mm frame for payload
Battery	4S 6,000 mAh LiPo (89 Wh, 450 g)	Larger than W2 to compensate for higher draw
Estimated hover power	~65 W (prop) + ~30 W (Jetson+LiDAR)	Total ~95 W
Estimated hover endurance	~35–45 min	Pending hardware validation
Heading accuracy (RTK)	<0.1° with dual-antenna 50 cm baseline	Required for cooperative localisation
Position accuracy (RTK)	<5 cm with RTK correction	Required for cooperative localisation
LiDAR range	70 m (Livox Mid-360 spec)	Worker tracking range
LiDAR FoV	360° × 59°	Full hemispherical below-horizon coverage
Cooperative loc. rate	≥20 Hz position injection to workers	Via ESP-NOW TDMA
Communication relay	900 MHz LoRa + 2.4 GHz ESP-NOW hub	Extends fleet comms range 2–3×

Table C1-1. Command node specifications.

C1 Component List

Component	Specification	Qty	Mass (g)	Cost ($)
Frame	Tarot TL65B44 or upgraded 700mm CFRP	1	360	120
Motor	T-MOTOR F90-1300KV (same as workers)	4	144	152
ESC	T-MOTOR F55A Pro II FOC	4	160	288
Propeller	HQProp 9×4.5 3-blade carbon	4+4	36	16
Flight Controller	Holybro Pixhawk 6C	1	60	110
Companion Computer	NVIDIA Jetson Orin NX 16GB	1	158	499
Jetson Baseboard	Holybro Pixhawk-Jetson baseboard	1	45	95
LiDAR	Livox Mid-360 360°×59° 200k pts/s	1	265	599
Depth Camera	Intel RealSense D435i	1	72	179
RTK GPS (primary)	Holybro H-RTK F9P Helical, dual-ant.	1	100	240

Component	Specification	Qty	Mass (g)	Cost ($)
RTK Antenna	50 cm CFRP baseline boom, 2× antennas	1	45	60
Long-range radio	Holybro SiK 900MHz 500mW	1	20	65
Mesh hub	ESP32 module (ESP-NOW hub firmware)	1	8	10
ELRS receiver	ExpressLRS EP2	1	5	18
Battery	4S 6,000 mAh LiPo 45C, 450 g	1	450	70
Power Module	Holybro PM02D isolated	1	12	29
Heatsink+Fan	Al 40×40 + 40 mm 5V fan	1	40	23
Wiring+Hardware	Full harness, fasteners	1 set	60	30
	TOTAL		**~2,040 g** electronics	**~$2,603**

Table C1-2. Command node component list.

C1 Cooperative Localisation Configuration

- **ROS2 node:** custom C1 coordination node runs on Jetson; subscribes to /livox/lidar and /camera/color; tracks worker IR ArUco markers; publishes /worker_positions at 20 Hz
- **Heading reference:** F9P dual-antenna provides absolute heading; set in PX4 as EKF2_AID_MASK to use dual-GPS heading
- **Worker ID:** each worker carries a unique ArUco marker (4×4 cm) at visible location; C1 vision pipeline maps ArUco ID to worker serial number
- **Position injection:** C1 packs [worker_id, x_global, y_global, z_global, timestamp] in ESP-NOW TDMA slot; worker injects as LANDING_TARGET at EKF external vision input
- **Latency budget:** LiDAR capture 5 ms + ArUco detect 8 ms + transform 2 ms + TDMA TX 5 ms + worker RX parse 2 ms = 22 ms total; within 40 ms dead-reckoning budget
- **Failover:** if C1 comms absent >10 s, workers auto-activate federated EKF via UWB and CBBA-SOC autonomous mode

K1 Ground Carrier Platform

Tier: Tier 3 — Logistics
Mass: ~25 kg
Endurance: Unlimited (mains power)

K1 is the Phase 1 carrier: a ruggedised ground vehicle providing two docking bays, battery management for 6 worker packs, communication relay, and self-mobility for repositioning up to 500 m from the ground station. It operates on mains power (via 50 m weatherproof cable) or 2× 24V 20Ah LiFePO4 on-board batteries for cable-free deployment up to 4 hours.

K1 Physical Layout

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ K1 TOP VIEW ■
■ ■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■ ■
■ ■ DOCK BAY #1 ■ ■ DOCK BAY #2 ■ ← D2 modules ■
■ ■ IR beacon 4LED ■ ■ IR beacon 4LED ■ ■
■ ■ pogo pins ×4 ■ ■ pogo pins ×4 ■ ■
■ ■ mag keel guide ■ ■ mag keel guide ■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ ■ ELECTRONICS BAY (weatherproof) ■ ■
■ ■ 2× Meanwell RSP-320-29.2 charger ■ ■
■ ■ 6× worker battery storage slots ■ ■
■ ■ ESP32 mesh hub + LoRa relay ■ ■
■ ■ 2× LiFePO4 24V 20Ah backup packs ■ ■
■ ■ STM32 carrier controller MCU ■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ ■
■ [Wheel FL] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ [Wheel FR] ■
■ ↕ motor + encoder (differential steer) ↕ ■
■ [Wheel RL] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ [Wheel RR] ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
Total footprint: 800 × 600 mm | Height: 400 mm | IP65 enclosure
```

K1 Component List

System	Component	Specification	Cost ($)
Dock Bay ×2	D2 Dock Module	IR beacon + pogo + mag guide (see D2 spec)	2 × $280 = $560
Charging	Meanwell RSP-320-29.2	29.2V/11A = 321W per bay × 2	2 × $85 = $170
Charging	BMS per worker pack ×6	4S active balancer, 100A	6 × $25 = $150
Onboard power	LiFePO4 24V 20Ah ×2	960 Wh, 15 kg total	2 × $200 = $400
Drive	DC brushed motor 12V 80W ×4	With 1:20 gearbox and encoder	4 × $35 = $140
Drive control	Roboteq MDC2230 dual-channel	60A, CAN, differential steer	$180
Navigation	u-blox M9N GPS + IMU	3 m accuracy self-nav	$85
Comms relay	900 MHz LoRa 500 mW	Extends GCS range to 8–10 km	$65
Comms hub	ESP32-S3 with dual Wi-Fi/BT	ESP-NOW mesh hub for fleet	$15
Controller	STM32H7 on custom carrier MCU board	Manages docking, BMS, drive, comms	$55

System	Component	Specification	Cost ($)
Enclosure	IP65 Al extrusion box 800×600×400	Hinged lid, cable glands	$180
Chassis	Al flat stock welded + anodised	Motor mounts, dock bay mounts	$120
Wiring, misc	XT90, marine connectors, cable glands	Weatherproof throughout	$80
	TOTAL K1		**~$2,200**

Table K1-1. K1 ground carrier component list.

K2 Airborne VTOL Carrier

Tier: Tier 3 — Mobile Logistics
Mass: ~5 kg MTOW
Endurance: 52 min (under load) / 120 min (own flight)

K2 is the Phase 2 airborne carrier. A heavy-lift hexacopter VTOL platform carrying two D2 docking bays and four worker battery reserve packs. It provides mobile battery logistics over the extended range of its own flight envelope. K2 allows the fleet to operate up to 5 km from the ground station without a return requirement for workers.

K2 Specifications and Component List

Component	Specification	Mass (g)	Cost ($)
Frame	Tarot T960 hexacopter CFRP folding, 960 mm	600	180
Motors ×6	T-MOTOR U8-II 85KV heavy-lift	6 × 220 = 1320	6 × 160 = 960
ESCs ×6	T-MOTOR Flame 80A FOC	6 × 65 = 390	6 × 95 = 570
Props ×6	T-MOTOR 18×6.1 carbon folding	6 × 90 = 540	6 × 35 = 210
FC	Holybro Pixhawk 6X (triple IMU)	100	350
K2 flight battery	6S 10,000 mAh Li-ion 450 Wh, 2.2 kg	2200	280
Worker battery packs ×4	4S 10,000 mAh Li-ion 148 Wh each	4 × 1100 = 4400	4 × 145 = 580
D2 dock bays ×2	IR beacon + pogo + mag keel guide	2 × 350 = 700	2 × 280 = 560
Chargers ×2	Compact 100W 29.2V/4A Li-ion charger	2 × 180 = 360	2 × 65 = 130
RTK GPS	Holybro H-RTK F9P + dual antenna	140	240
Comms relay	900 MHz LoRa 500 mW + ESP32 mesh	30	80
K2 controller	STM32H7 custom carrier MCU	25	55
Structure	Custom Al/CFRP dock bay cradle	600	200
Wiring + misc	XT90, power rails, cable management	150	80
	K2 TOTAL (excl. worker batteries)	**~5,155 g**	**~$3,855**
	K2 TOTAL (with 4 worker batteries)	**~9,555 g**	**~$4,435**

Table K2-1. Airborne carrier component list. MTOW constrained to 5 kg by removing 2 of 4 worker batteries for shorter sorties.

K2 hover power estimate: P = 0.639 × (5.0 × 9.81)^1.5 ≈ 180 W (propulsion). With 2 bays recharging at 2×100W: total draw ≈ 380 W. At 450 Wh flight battery (80% DoD = 360 Wh): endurance under full recharge load = 360/380 × 60 ≈ 57 min.

D1 Fixed Docking Station

Tier: Tier 3 — Infrastructure
Mass: ~12 kg installed
Endurance: N/A (mains connected)

D1 is a standalone fixed ground docking station providing one high-bandwidth docking bay with IR beacon guidance, magnetic alignment, pogo-pin power transfer, and data upload. It connects to mains power (or 48V DC bus) and provides weather-protected worker recharge at a fixed deployment point. D1 may be pole-mounted, wall-mounted, or ground-mounted.

D1 Functional Diagram

```
[WORKER DRONE APPROACH ↓]
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ IR BEACON ARRAY ■
■ LED■(11kHz) LED■(17kHz) ■
■ LED■(23kHz) LED■(31kHz) (940 nm) ■
■ Detection range: 2 m, ±60° cone ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ drone descends on ε error
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ MAGNETIC KEEL FUNNEL ■
■ Al V-funnel: corrects ±50 mm lateral ■
■ N52 magnets (×4, 4 kg pull each) ■
■ Holding force: 64 N minimum ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ keel engages
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ POGO-PIN CONTACT ARRAY ■
■ 4 pins: BAT+, BAT–, DATA+, DATA– ■
■ Rating: 10A continuous, gold-plated ■
■ Resistance: <0.1 Ω contact ■
■ SSR: no current until R-match detect ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ CHARGE CONTROLLER ■
■ 29.2V / 4A CC-CV (Li-ion compatible) ■
■ OR 16.8V / 4A (LiPo compatible) ■
■ BMS monitoring: cell balance, temp ■
■ Mains input: 100-240V AC, 50/60 Hz ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

D1 Component List

Component	Specification	Cost ($)
IR Beacon PCB	Custom: 4× OSRAM SFH4557 LEDs + STM32G031 driver	45
IR Beacon Enclosure	Polycarbonate IP65 dome, 80 mm diameter	15
Magnetic Funnel	CNC Al 6061 V-funnel, 4× N52 10×3 mm magnets embedded	95
Pogo Pin Assembly	4× gold-plated 10A spring-loaded pins + PCB	40
SSR Contact Gate	Solid-state relay 40A; opens only on R-match confirm	25
Charge Controller	Genasun GVB-8 or custom 29.2V/4A CC-CV with BMS	85

Component	Specification	Cost ($)
Structure	Powder-coated steel base plate, mounting options: ground/pole/wall	65
Weatherproof enclosure	IP65 electronics box with cable glands	45
Cable assembly	10 m weatherproof DC cable to mains PSU	25
Mains PSU	Meanwell LRS-350-29, 29V/12A = 348W, DIN-rail	55
	D1 TOTAL (one bay)	**~$495**

Table D1-1. Fixed docking station component list.

D2 Carrier-Mounted Docking Module

Tier: Tier 3 — Infrastructure
Mass: ~350 g
Endurance: N/A (powered by K1/K2)

D2 is the compact docking module integrated into K1 and K2 carriers. It is identical in function to D1 but built as a self-contained module that mounts into the carrier bay and receives 29V DC from the carrier charge bank. Two D2 modules per K1, two per K2 (standard). D2 includes a 2-axis compliant mount to absorb vibration on the airborne K2 carrier.

Component	Specification	Mass (g)	Cost ($)
IR Beacon PCB (integrated)	Custom 60×60 mm with 4× SFH4557 + STM32G031	18	38
Funnel + magnet assembly	Compact Al funnel 120×120×60 mm with N52 magnets	180	65
Pogo pin assembly	4-pin gold-plated PCB module	15	30
SSR module	40A SSR with R-match circuit on-board	20	22
Vibration isolation (K2 only)	4× rubber isolators 20×20 mm on base	30	12
Module housing	CFRP/Al plate, 150×150×90 mm, IP54	85	45
Internal wiring	JST-XH power + UART to carrier controller	8	8
	D2 TOTAL per module	**~356 g**	**~$220**

Table D2-1. Carrier-mounted dock module component list.

COMPONENT SELECTION GUIDE & SUBSTITUTIONS

Motor Selection Criteria

Motors for the worker class must satisfy: max thrust ≥2.8 kg per motor (×4 = 11.2 kg for 1.97 kg drone with 5.7× thrust-to-weight); KV rating matching voltage (4S = 14.8V nominal; KV × V = hover RPM ~5,000 target); stator diameter ≥45 mm for adequate torque; motor weight ≤40 g per unit.

Motor	KV	Weight	Max Thrust (9in)	Cost	Status
T-MOTOR F90-1300KV	1300	36 g	2.8 kg @ 10k RPM	$38	PRIMARY (validated)
T-MOTOR F80-1900KV	1900	34 g	2.4 kg @ 9k RPM	$35	Alternative (9in marginal)
EMAX ECO II 2807-1300KV	1300	58 g	2.6 kg	$22	Budget option (+22 g/motor)
T-MOTOR Velox V2306 1750KV	1750	30 g	2.1 kg @ 10k RPM	$28	Too low thrust at hover

Table CS-1. Motor alternatives. Primary validated; alternatives require re-calibration.

ESC Selection Criteria

FOC capability is mandatory (not optional) for the vibration reduction benefit. Required: FOC with sinusoidal commutation at ≥16 kHz switching frequency; current sensing capability for telemetry; >50A continuous current rating for headroom at max throttle; BLHeli_32 or proprietary FOC firmware with DSHOT600 support.

ESC	Current	FOC	Cost	Status
T-MOTOR F55A Pro II	55A cont.	Yes, BLHeli_32 FOC	$72	PRIMARY (validated)
Hobbywing Platinum 60A FOC	60A cont.	Yes, proprietary FOC	$68	Alternative — good
AM32 open-source FOC 50A	50A cont.	Yes, AM32 FOC	$30	Budget — limited validation
Generic BLHeli_32 40A	40A cont.	No — BLDC only	$15	REJECTED — no FOC benefit

Table CS-2. ESC selection. FOC mandatory; generic BLDC ESCs do not achieve the documented vibration reduction.

Battery Chemistry Decision Matrix

Chemistry	Energy Density	Max C-Rate	Cycle Life	Use Case	Verdict
LiPo (current W2)	150–180 Wh/kg	45C peak	200–300 cycles	W2 validated; high C for manoeuvres	W2 primary
Li-ion 18650 (W1 target)	200–240 Wh/kg	5–10C max	500–1000 cycles	W1 endurance-optimised	W1 primary
Li-ion 21700	220–260 Wh/kg	5–8C max	500–800 cycles	Higher energy W1 variant	Future W1
LiFePO4	120–160 Wh/kg	5C cont.	2000+ cycles	K1 carrier backup banks	K1 storage
Solid-state (target 2027–28)	400–500 Wh/kg (projected)	TBD	TBD	Double endurance future workers	Phase 3 watch

Table CS-3. Battery chemistry comparison and usage assignments.

Flight Controller Selection

All units use Holybro Pixhawk 6C as primary flight controller. The STM32H743 MCU (480 MHz, M7) provides adequate compute for 400 Hz attitude loop, EKF3 fusion, and custom IR landing target plugin. The ICM-42688-P IMU is validated to <0.5 mm/s vibration RMS at hover with FOC+CFRP — meeting the SOC estimation accuracy requirement. Alternative: CubePilot Cube Orange+ (higher cost, triple-redundant IMU; recommended for Phase 2 command node if C1 criticality increases).

SYSTEM INTEGRATION QUICK-REFERENCE

Interface	Protocol	Speed / Rate	Units Connected
FC ↔ ESC	DSHOT600	600 kbps	All flight units
FC ↔ GPS	UART 115200 baud	10 Hz position	All flight units
FC ↔ Companion (Jetson)	MAVLink 2.0 over UART 921600 baud	100 Hz	W2, C1
Jetson ↔ LiDAR	Ethernet UDP	200k pts/s	C1 only
Jetson ↔ Depth Camera	USB 3.0	30 fps	W2, C1
FC ↔ IR board	UART 500 Hz	[ε_x, ε_y, h] struct	W1, W2, C1
FC ↔ UWB	SPI 8 MHz	Ranging at 50 Hz	W1, W2
FC ↔ Power Module	I²C (PM02D)	10 Hz V/I/SOC	All flight units
Drone ↔ Dock	Pogo pins (power)	10A DC	All workers
Drone ↔ Dock	Pogo pins (data)	USB 2.0 / UART	All workers (log upload)
Fleet ↔ Fleet	ESP-NOW 2.4 GHz	<5 ms, 250 kbps shared	All units
Fleet ↔ GCS	LoRa 900 MHz	20–50 ms, 19.2 kbps	C1 → GCS

Table SI-1. System interface quick-reference.